

Title

“Drawing Intelligence: Sketching as a Computational Language”

Rhetorical Argument

In design, drawing is the primary human natural language. Because it is a highly mimetic medium, meaning it closely mimics the thing it is trying to represent, it is perfectly suited to communicate complex spatial information—such as 3D forms, shapes, spatial locations, textures, and non-linear workflows and instructions. Therefore, drawing must be formally recognized and integrated as a computable language for human-computer interaction. The necessity for this shift becomes clear when examining current AI applications, which are dominated by the linear chatbot interfaces popularized by ChatGPT. While these systems allow for one-dimensional communication via text prompts and file uploads, they are fundamentally insufficient for accurately articulating complex, three-dimensional logic and spatial instructions. Just as AI today is being used throughout the software industry to help developers write code, a similar revolution is overdue in design and engineering. Currently, navigating the deep technicalities of CAD platforms consumes a significant amount of time, often pulling focus away from the actual design problem at hand. By integrating drawing as a computable language, the designer gains the option to instruct an AI to handle the tedious mechanics of these complex digital environments, freeing the professional to stay more focused on the creative act of designing.

Research question(s)

- Where is drawing situated among the many ways in which we communicate and store information?
- What are the different kinds of drawing, how can they be categorized?
- How can we use drawing as a means of communication and interaction with AI and other digital technologies?
- What are the ways in which we interact with computers today? What information are we able to communicate and what are we unable to communicate?
- How is drawing used in various design practices?
- What are the problems faced in the many fields of design?
- What are the various types of digital drawing, how should UIs and devices be designed for them?

Keywords (5)

Drawing, Language, Communication, Design, AI, Human-Computer Interaction

Intersecting Fields

Linguistics, Design, Human-Computer Interaction, AI

Historical Lineage

- History of Human-Computer Interaction, Computer input and output devices, Languages used to communicate with computers.
- History of use of drawing in art, design, communication, information storage.
- History of work in design fields (Architecture, Products Design, Animation, Engineering etc.)

Your community of practice, Situated technology

Books, Essays, Papers:

- Linkography: Unfolding the Design Process (Book) — Gabriela Goldschmidt - Maps out the syntax of creative thought by examining the continuous feedback loop between a designer's mental imagery and their physical marks on paper.
- Sketches of Thought (Book, 1995) — Vinod Goel - Argues that the inherent ambiguity of freehand sketching is essential for cognitive flexibility, framing the sketchpad as a physical extension of human working memory.
- The Alphabet and the Algorithm (Book) & "The Art of Drawing" (Essay) — Mario Carpo - Examines the historical evolution of design logic, contrasting the traditional limitations of geometric drawing with the modern capabilities of algorithmic computation.
- Translations from Drawing to Building (Book/Essay) — Robin Evans - Explores how architectural drawing is not a neutral representation of an idea, but an active, restrictive language that dictates what can actually be materialized.
- Points + Lines: Diagrams and Projects for the City (Book) — Stan Allen - Frames architectural drawings and diagrams as operative notations—functioning much like a musical score to map out dynamic relationships and behaviors rather than static pictures.

Active Projects:

- Rhino 9 Beta Markup: Draw, callout and annotate directly on the view using the tools provided - <https://discourse.mcneel.com/t/rhino-beta-feature-markup/204955>
- Claude Design: Prompting through sketches. Beyond standard text inputs, the platform now features a drawing tool that allows users to guide the AI by sketching directly over a UI canvas.

Past Projects:

- RoboSketch - by a team of researchers from KAIST (Korea Advanced Institute of Science and Technology) - Draw in a 3D viewport and animate the resulting skeletons to make robots. - SIGGRAPH Asia 2024 - https://www.youtube.com/watch?v=5wi53Z2_sAk
- SILK: Sketching Interfaces Like Krazy - James A. Landay PhD at Carnegie Mellon University (CMU) - sketch user interface elements by hand and make them interactive. <https://dl.acm.org/doi/pdf/10.1145/257089.257396>
- Stable Diffusion: ControlNet - generates detailed images from drawings by identifying lines in the sketches using AI. - <https://stablediffusionweb.com/ControlNet>
- DrawTalking - Karl Toby Rosenberg - enables users to animate scenes by speaking commands linked to sketches. - <https://ktrosenberg.github.io/drawtalking/>

Methods

- Types of 2D, 3D drawing in CAD software - 3D “pipe” drawing, 2D raster sketching to planes in 3D viewport, Polyline CAD drawing, Spline CAD drawing.
- My project will combine drawing as a language related theory as well as a software prototype. (not now, but for my final capstone in upcoming semesters)

Computational Design Experiments

I am working on an app that is an AI agent/assistant for CAD software. In this app I have made a drawing tool that can be used inside a 3D viewport. For my Capstone project my focus will only be on the drawing tools and the drawing UI of the AI agent.

Visual Representation

I will use a sketchbook like visual style for my presentation website. It will have sketches, images and text mixed together to explain concepts and information, just like we would make notes in our personal notebooks.

The Challenge

- Coding - Making a working prototype software, including the various types of drawing tools and user interfaces.
- Collaboration - Forming a team of collaborators who are interested in working with me on this project.
- Data Structures - Exploring different ways of structuring the drawing data into a numerical format ideal for a neural network.

A printed vertical drawing

A drawing showing the lineage of the history of human-computer interaction, input and output language and devices.

A material-spatial gesture

Digital 3D graph with different types of human language plotted. Interactive game where users can get a card with a specific language type on it, and they need to place it in the 3D graph. The screen will be projected onto a wall.